

Performance Testing Report (PTR)

Date	03 October 2026
Developer / Student	Vinay Krishna A K
Project Name	Automated Network Request Management in ServiceNow

Project Overview:

Project Name: Automated Network Request Management in ServiceNow

Project Description: Performance and throughput validation of the self-service Network Request catalog item (/sp), client-side dynamic UI policy evaluations, 10-step Flow Designer execution latency, and Database Tables (u_database_table) CRUD transaction speeds on the Personal Developer Instance (PDI).

Project Version: Release v1.0 (Production-Ready Build)

Testing Period: 21 September 2026 to 03 October 2026

Testing Scope:

- Service Portal (/sp) initial page load time and widget rendering.
- Catalog UI Policy client-side response time when toggling Type of Connection (New/Existing) and Mode of Payment (UPI/Card).
- Flow Designer end-to-end execution latency across all 10 steps (Variables ingestion, Record creation, Email dispatches, Approval routing).
- Database Tables (u_database_table) insertion and state update throughput under single and concurrent request simulations.
- Outbound email generation and delivery queue latency (Steps 3, 7, and 10).
- Platform ACL overhead and table query performance.

Testing Environment:

- Instance URL: ServiceNow Cloud Personal Developer Instance (/sp)
- Network Conditions: Simulated 4G (15 Mbps) and High-Speed Broadband (100 Mbps)
- Test Client: Google Chrome DevTools (Network & Performance Profiler), Glide Transaction Logs, ServiceNow Flow Execution Details

Performance Test Cases & Benchmarks:

Test Case ID	Test Scenario / Workload	Test Steps / Execution Procedure	Expected Benchmark (SLA)	Actual Result Measured	Pass / Fail
PTC-001	Portal Catalog Item Load Latency	1. Clear browser cache. 2. Navigate to /sp catalog URL. 3. Open 'Network Request' item. 4. Measure DOM Complete & Load Event via DevTools.	Page DOM load time < 2.0 seconds on standard broadband connection.	Average Load Time: 1.18s; DOM Content Loaded: 0.74s.	Pass
PTC-002	Client UI Policy Dynamic Field Evaluation	1. Open Network Request form on /sp. 2. Toggle Type of Connection between 'New' and 'Existing'. 3. Switch Mode of Payment between 'UPI' and 'Card'. 4. Profile client script execution time.	Dynamic field reveal/hide execution latency < 100 ms with zero perceptible lag.	Field state transition completed in 32 ms.	Pass
PTC-003	Custom Table Insertion Speed (Step 2)	1. Fill mandatory fields (Requested For, Mobile, Connection Type, Fee, Mode, Address). 2. Click Submit on portal. 3. Measure transaction duration to create record in u_database_table.	New record insert + auto-number allocation < 1.0 second.	Record inserted in 380 ms with sequential request auto-number.	Pass
PTC-004	Submission Acknowledgment Email Latency (Step 3)	1. Complete form submission on /sp. 2. Flow Designer Step 3 dispatches acknowledgment email. 3. Check sys_email table queue time.	Outbound email generated and queued within 3.0 seconds of submission.	Email generated and placed in 'ready' queue in 1.45s.	Pass

PTC-005	Approval Gate Action Latency (Step 4)	1. Flow triggers Step 4: Ask For Approval. 2. Measure creation of sysapproval_approver record and related list update.	Approval request record created and linked to u_database_table in < 1.5 seconds.	Approval task generated and linked in 620 ms.	Pass
PTC-006	Conditional Branch & Status Update Latency (Steps 5-7)	1. Approver clicks 'Approve' on approval page. 2. Flow Step 5 evaluates condition true. 3. Measure Step 6 record status update and Step 7 email notification trigger.	Full branch execution and record state update completed in < 2.5 seconds.	Record updated to 'Approved' in 890 ms; approval email queued in 1.20s (Total: 2.09s).	Pass
PTC-007	Rejection Branch & Notification Latency (Steps 9-10)	1. Approver clicks 'Reject'. 2. Flow Step 5 evaluates false. 3. Measure Step 9 record state update and Step 10 rejection email trigger.	Rejection branch execution and record state update completed in < 2.5 seconds.	Record updated to 'Rejected' in 840 ms; rejection email queued in 1.15s (Total: 1.99s).	Pass
PTC-008	Attachment Upload Throughput	1. Attach a 5 MB PDF document to 'Add attachments' widget. 2. Submit request. 3. Measure upload and sys_attachment chunking time.	5 MB attachment uploaded and processed in < 3.0 seconds.	File uploaded and indexed in sys_attachment in 1.82s.	Pass
PTC-009	End-to-End Flow Designer Execution Throughput	1. Execute complete 10-step flow from Service Catalog trigger to completion. 2. Verify Flow Execution Details duration log.	Total automated flow execution duration < 5.0 seconds (excluding manual approval wait).	Total system compute duration: 3.42 seconds across all 10 active steps.	Pass

Performance Bottleneck & Optimization Tracking:

Issue ID	Performance Bottleneck / Issue Description	Steps to Reproduce	Severity	Status	Optimization / Resolution
PB-001	Initial Service Portal widget delay due to uncached client script dependencies.	1. Open incognito window. 2. Navigate to /sp. 3. Measure Time to First Byte (TTFB).	Medium	Closed	Enabled platform widget caching (GlideCache) and removed redundant console statements; load time reduced by 42%.
PB-002	Sequential email generation in Flow Designer causing 1.2s queue delay during peak test runs.	1. Trigger multi-request submission. 2. Inspect Flow Execution step timestamps.	Low	Closed	Configured email dispatches to execute asynchronously via platform event queue; UI remains immediately responsive.
PB-003	Unindexed query scan on u_database_table when loading approval related lists.	1. Open approval record. 2. Monitor SQL transaction response time.	Low	Closed	Added database B-Tree index on u_database_table reference and state fields; query response improved to < 40 ms.

Sign-off:

Performance Lead / Student Tester: Vinay Krishna A K

Date of Test Completion: 03 October 2026

Test Verdict: PASSED — All latency, throughput, and execution SLAs met with 100% compliance across 9 test scenarios.